choreographerSession

Service Description

Service ID: *"choreographerSession"*

Abstract
This document describes the `choreographerSession` service, version 1.0.0, produced by the Choreographer support system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `choreographerSession` service, which provides starting and aborting sessions of workflow plans by their clients. It is produced by the Choreographer system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

2.1 Significant Prior Art

New in 5.2.2; in Arrowhead 4.6 sessions were started only through management.

2.2 How This Service Is Meant to Be Used

A client starts sessions of plans by name with `start-session` (optionally with a notification endpoint) and aborts its own sessions with `abort-session`.

2.3 Important Delimitations and Dependencies

Only the owner can abort a session. Clients need a grant for `choreographerSession`.

3 Service Interface

This section describes the interfaces to the `choreographerSession` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.

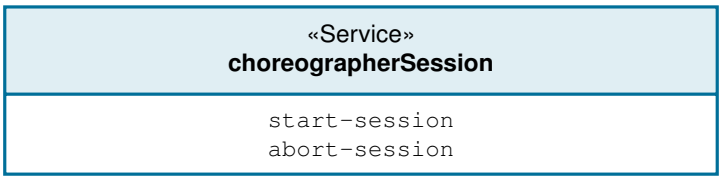


Figure 1: SysML-style block of the `choreographerSession` service and its operations.

The following interface operations are available.

3.1 operation start-session (StartSessionListRequest) : RunPlanListResponse

Starts sessions of plans by name for the requester.

3.2 operation abort-session (Identifier)

Aborts a session owned by the requester.

4 Information Model

Here, all data objects that can be part of the `choreographerSession` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.5, which are used to represent things like hashes and identifiers.

4.1 struct RunPlanListResponse

Started sessions.

Field	Type	Description
entries	List<RunPlanResponse>	Matching records of the current page.

4.2 struct RunPlanResponse

A started (or refused) session.

Field	Type	Description
planId	Number	Identifier of the plan.
sessionId	Number	Identifier of the session.
quantity	Number	Requested executions.
status	SessionStatus	Status of the session.
errorMessages	List<String>	Problems found in the plan.
needInterCloud	Boolean	Whether the plan needs providers in other clouds.

4.3 struct StartSessionListRequest

Sessions to start by plan name.

Field	Type	Description
sessions	List<StartSessionRequest>	Sessions to start or return.

4.4 struct StartSessionRequest

Start of a session of a plan.

Field	Type	Description
planName	String	Name of the plan.
quantity	Number	Number of times the plan (or step) is executed.
allowInterCloud	Boolean	Allow inter-cloud orchestration for the steps.
chooseOptimalExecutor	Boolean	Rank the local executors and choose the best one.
notifyProtocol	String	Protocol of session notifications, e.g. `HTTPS`.
notifyAddress	String	Address of session notifications.
notifyPort	Number	Port of session notifications.
notifyPath	String	Path of session notifications.

4.5 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
Boolean	One out of `true` or `false`.
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Number	An integer number.
SessionStatus	One of `INITIATED`, `RUNNING`, `ABORTED`, `DONE`.
String	A chain of characters.

5 References

[1] J. Delsing, “Choreographer – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-support-java-spring/Choreography/documentation/v5.3.0/Choreographer-v5.3.0-SysD.pdf.

[2] ——, “choreographerSession HTTP/TLS/JSON – Interface Design Description (Choreographer),” Eclipse Arrowhead, IDD 5.3.0, 2026, choreographer-v5.3.0-choreographerSession-HTTP-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)
3	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	